

TaskFlow

Enterprise Task Management System

Group details:-

Group Members:

S. No.	Student Name	Registration Number	Application Number
1.	PV Rohan	23BCE11428	IN26012491
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MPOne - Advanced Software Engineering & Development Internship Batch 10(B) - Snehal Wani (6:00-8:00 pm)

Code github repository

<https://github.com/ROHAN-3105/Task-managament>

TECHNOLOGY STACK:

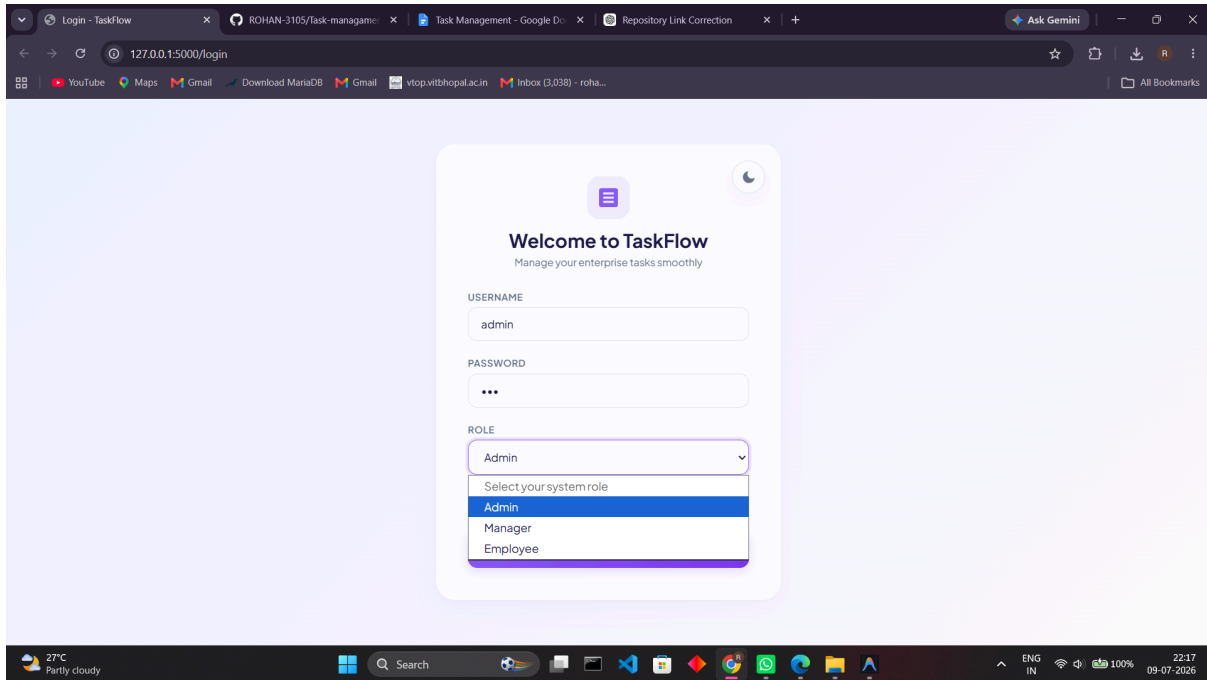
- Backend: Python Flask, MySQL Connector
- Frontend: HTML5, CSS3 (Custom Glassmorphism), Vanilla JavaScript, [Chart.js](#)
- Database: MySQL with relational schema and integrity.

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1. Secure Login & Authentication

TaskFlow features a secure, role-based authentication system that allows users to log in as an **Admin**, **Manager**, or **Employee**. Passwords are securely hashed before being stored in the database using **Werkzeug's password hashing algorithms (bcrypt/PBKDF2)**, ensuring strong security. The user interface incorporates a modern glassmorphism design with premium CSS variables, smooth animations, and interactive input elements to provide an enhanced user experience.



Key Features Verified on this Screen:

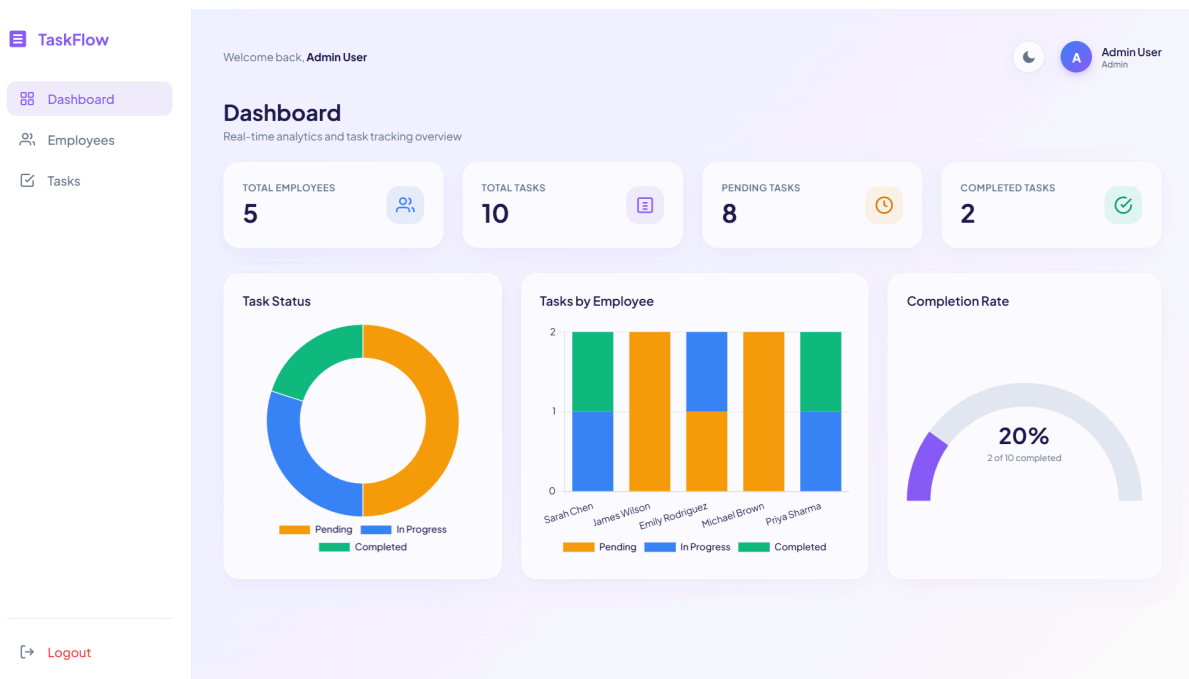
- Dynamic role selection through a dropdown menu (Admin, Manager, Employee).
- Front-end form validation combined with backend authentication and informative error messages.
- Secure session management using persistent session cookies with role-based access control and automatic redirection after successful login.

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2. Real-time Admin/Manager Dashboard

After successful authentication, administrators and managers are redirected to a real-time analytics dashboard. The dashboard utilizes **Chart.js** to display dynamic visualizations of task distribution, employee workloads, and task completion statistics. It features a fully responsive layout built using CSS Grid, modern dashboard cards, and interactive hover effects for an intuitive user experience.



Key Features Verified on this Screen:

- Real-time analytics displaying **Total Tasks**, **Pending Tasks**, **Completed Tasks**, and **Total Employees**.
- Dynamic **Chart.js** visualizations illustrating task status distribution and employee workload.
- High-performance MySQL database queries performing aggregation operations to generate real-time dashboard statistics efficiently.

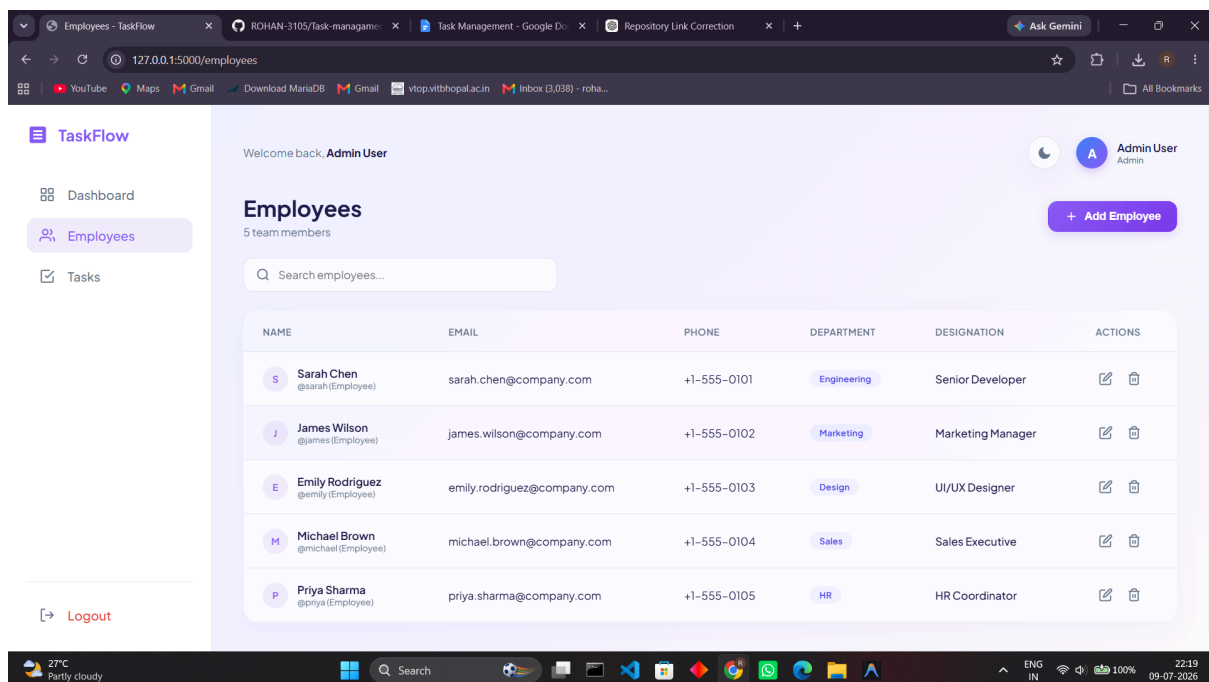
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3. Employee Management Module

The **Employee Management Module** enables administrators to efficiently manage employee records through a modern, responsive interface. The module provides complete **CRUD (Create, Read, Update, Delete)** functionality, allowing administrators to add new employees, update existing employee information, and remove inactive records. A real-time search feature enables quick filtering of employees, while department and designation details are displayed in a clean tabular layout for easy access.

The interface is designed using a premium glassmorphism-inspired theme with responsive cards, smooth animations, and intuitive action buttons, ensuring an enhanced user experience across different screen sizes.



Key Features Verified on this Screen:

- Complete **CRUD operations** for employee records (Add, Edit, Delete).
- Real-time employee search with instant filtering.
- Displays employee information including **Name, Email, Phone Number, Department, and Designation**.
- Department-wise categorization using visually distinct badges.
- Responsive table layout with modern UI components and hover effects.
- Role-based access control ensuring only authorized users can manage employee records.
- Seamless integration with the MySQL database for secure and efficient data retrieval and updates.

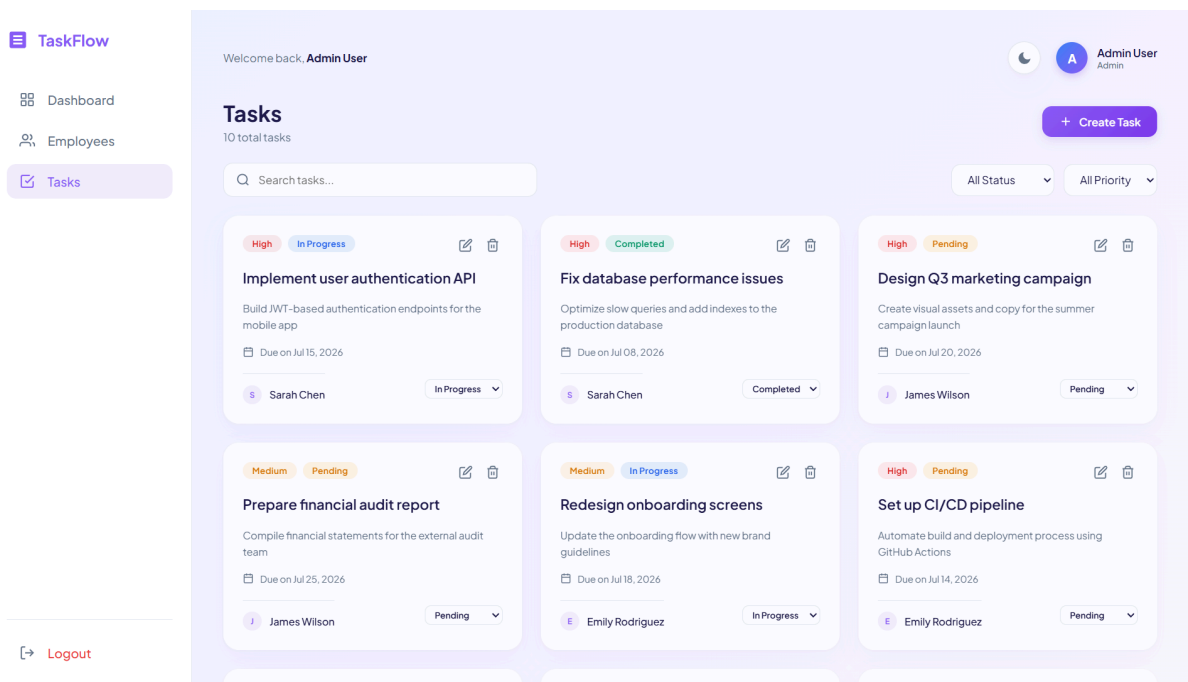
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4. Task Management Module

The **Task Management Module** enables administrators and managers to efficiently create, assign, monitor, and update tasks through an interactive card-based interface. Each task card displays essential information, including task title, description, priority level, assigned employee, due date, and current status. The module supports real-time task tracking, allowing users to update task progress seamlessly while maintaining accurate records in the MySQL database.

The interface is designed using a modern glassmorphism theme with responsive layouts, smooth animations, and intuitive controls, providing an efficient and user-friendly task management experience.



Key Features Verified on this Screen:

- Create, edit, and delete tasks through dedicated action buttons.
- Assign tasks to employees with configurable priority levels (High, Medium, Low).
- Real-time task status updates (Pending, In Progress, Completed).
- Search functionality for quickly locating specific tasks.
- Filter tasks based on **Status** and **Priority**.
- Displays task details including title, description, assigned employee, and due date.

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